

AUSTPIC Presents Intra AUST Programming Contest - Spring 2025 (Preliminary Round)

Problem Set

Coder0J

A. RedX Again

Time: 1 s

Memory: 128 MB

The long night has fallen. The White Walkers and their Army of the Dead have finally reached the grounds of RedX. The students of RedX are scattered across the tiled courtyard. Fortunately, the Children of the Forest have cast a powerful warding spell over the grounds. This spell strictly prevents the wights (the Army of the Dead) from touching anyone.

However, the spell offers no protection against the White Walkers. The Night King (leader of White Walkers) stands perfectly still at a specific coordinate (X, Y) and carries exactly M magically imbued ice spears. He will throw all of his M spears into the courtyard one by one, choosing a direction for each. Once thrown, a spear travels infinitely along a straight trajectory, instantly eliminating any student in its path.

Given the coordinates of the students, the Night King's location, and the M spears he must throw, your task is to determine the maximum number of students eliminated in a single throw.

Input

The first line of the input contains a single integer T ($1 \leq T \leq 10$), the number of test cases.

For each test case:

The first line contains two space-separated integers N and M ($1 \leq N, M \leq 10^3$), representing the number of students and the number of spears, respectively.

The second line contains two space-separated integers X and Y ($-10^9 \leq X, Y \leq 10^9$), representing the exact coordinate of the Night King.

The next N lines each contain two space-separated integers x_i and y_i ($-10^9 \leq x_i, y_i \leq 10^9$), representing the coordinates of the i^{th} student.

It is guaranteed that no student is standing at the exact same coordinate as the Night King, and no two students share the exact same coordinate.

Output

For each test case, print a single integer on a new line: the maximum number of students eliminated in a single throw.

Examples

Input	Output
2	3
5 10	4
0 0	
1 1	
2 2	
3 3	
1 4	
4 1	
6 3	
0 0	
1 1	
2 2	
3 3	
4 4	

Input	Output
-1 -1 -2 -2	

B. Lichu Chor

Time: 10 s

Memory: 256 MB

যাব ফের? কেন মলি ভাই,
চুরিতে আর যদি যাই!
তবে মোর নামেই মিথ্যা!
লিচু চোর — কাজী নজরুল ইসলাম

Khokon wants to steal all the lichu from a bag in *kele da's Bagan*. Initially, the bag contains N distinct lichu, numbered from 1 to N . Khokon performs a sequence of moves until the bag becomes empty.

Let r be the number of lichu currently remaining in the bag. In one move, Khokon selects one subset uniformly at random from all 2^r subsets of the remaining lichu's. However, M of these subsets are illegal and cannot be used.

- If the selected subset is illegal, the move has no effect.
- Otherwise, if all lichu in the selected subset are still present in the bag, Khokon steals all of them simultaneously, removing them from the bag.

Khokon repeats this process until there are no lichu left. Compute the expected number of moves required for Khokon to steal all the lichu.

Input

The first line contains two integers N ($1 \leq N \leq 18$) and M ($0 \leq M$) — denoting number of lichu's and the number of illegal subsets.

Each of the next M lines describes one illegal subset.

The i^{th} line starts with an integer K ($1 \leq K \leq N$), followed by K distinct integers $L_1, L_2, \dots, L_k$ ($1 \leq L_i \leq N$) the indices of lichu in this illegal subset.

All illegal subsets are distinct.

Output

If it is impossible to empty the bag, print a single line : -1. Otherwise, print a single integer — the expected number of moves as a fraction modulo 998244353.

Formally, Let the expected number of moves be an irreducible fraction $\frac{p}{q}$.

Output $p \cdot q^{-1}$ modulo 998244353, where q^{-1} is the modular inverse of q modulo 998244353.

Examples

Input	Output
2 1 2 1 2	4

C. Zoro's Grand Tour of AUST

Time: 1 s

Memory: 128 MB

Zoro has been trying to find the IAPC Spring '25 event venue for three hours. He's passed the same tea stall four times. His sense of direction, famously the worst in any ocean or land, has completely failed him once again.

Finally, fed up and slightly embarrassed, he pulls out his den den mushi and calls Nami, *"Just tell me where to go!"*

Nami sighs, *"I KNEW this would happen."*

She can't draw a proper map over the phone, so she does the next best thing. She describes a grid of 26 landmarks on AUST campus, arranged in two rows, left to right:

Nami's Campus Grid

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

She tells him: *"Start at the very first landmark on the top row. Follow my directions exactly. Every place you stop at press it into your memory. The sequence of places you visit, in order, spells out exactly where you need to go."*

Zoro grunts. *"Fine. Talk."*

Nami's Directions (follow in order):

1. You start at the *first landmark, top row*.
2. Go *7 steps right*, then *1 step down* to the bottom row. Stop. Mark it.
3. Go *2 steps left*. Stop. Mark it.
4. Go *1 step right*. Stop. Mark it.
5. Go *4 steps left*. Stop. Mark it.
6. Go *6 steps right*, then *1 step up* to the top row. Stop. Mark it.
7. Go *6 steps left*. Stop. Mark it.

"Now the 7 landmarks you marked, in the order you visited them that's your destination. Figure it out, Zoro."

Zoro stared at the grid for a full minute.

"I still don't get it," he muttered.

She angrily says, *"JUST FOLLOW THE NUMBERS, ZORO!"*

He did. Eventually. And he found the place.

Can you?

Input

There is no input. The story above is the entire test case.

Output

Print the destination name in uppercase.

D. Game of All

Time: 1 s

Memory: 128 MB

Sajid is playing the video game "Game of All" in which his character is trapped in a deadly maze. To exit the maze and advance to the next level, he must solve a riddle.

He receives a number of balloons, each bearing a printed number. The balloons are placed in a certain order. The written number can be represented by $A_1, A_2, A_3, \dots, A_n$.

He must select a few balloons, and he must perform various operations between values. While keeping the order of the indexes, the operations are continually carried out in sequence.

The sequence of operation are

1. Plus (+)
2. Bitwise AND (&)
3. Bitwise OR (|)
4. Bitwise XOR ($\oplus$)

This order of operation (*Plus*, *Bitwise AND*, *Bitwise OR*, and *Bitwise XOR*) will hold even if we choose more numbers. He can stop the operation at any point.

Basically if the fourth operation is *Bitwise XOR*, then the fifth operation will be *Plus*, then sixth operation will be *Bitwise AND* and so on.

For example, suppose he is given balloons with the values:

4, 5, 6, 7, 8, 9, 10, 12, 18, 24, 28

He may perform operations in the following manner:

- 4
- (4 + 5)
- ((4 + 6) & 7)
- (((4 + 6) & 7) | 8)
- (((((4 + 6) & 7) | 8) $\oplus$ 10) + 18)
- ((((((5 + 6) & 8) | 9) $\oplus$ 12) + 18) & 28)

A sequence is considered valid only if:

1. The balloons are selected in increasing index order (their relative order in the original list must be preserved).
2. Operations are applied sequentially, where each new operation uses the result of the previous expression and the next selected balloon.

Examples of invalid expressions:

- (7 + 4) and (7 + 6) — invalid because the order of indices is not preserved.
- ((4 & 6) + 8) — invalid because the required sequence of operations is not maintained.

Sajid is unable to solve the problem where he needs to maximize the value. So he asks you to solve the problem so that he can solve the level.

Print the maximum value.

Input

The first line of the input contains a single integer t ($1 \leq t \leq 10$) — the number of test cases.

Each test case consists of two lines:

The first line contains a single integer n ($1 \leq n \leq 20$) — the number of balloons.

The second line contains n integers $A_1, A_2, \dots, A_n$ ($1 \leq A_i \leq 10^9$) — the original sequence of balloons.

It is guaranteed that the sum of n over all test cases does not exceed 20.

Output

Print maximum value for each test case.

Examples

Input	Output
2	11
3	28
4 5 6	
4	
4 5 18 10	

Notes

In Test Case 1, the expression $(5 + 6)$ evaluates to 11, which is the maximum possible value. It is greater than or equal to the results of all other valid expressions, such as $((4 + 5) \& 6)$, $(4 + 6)$, $(4 + 5)$, and 5. In fact, no other valid sequence of operations can produce a value larger than 11. Therefore, the answer for Test Case 1 is 11.

In Test Case 2, the expression $(18 + 10)$ evaluates to 28, and no other valid expression in the given sequence can produce a value greater than 28. Hence, the maximum achievable value is 28, which is the answer for Test Case 2.

E. Street Lights and Mex

Time: 1 s

Memory: 128 MB

After surviving a grueling day-long programming contest, *HRK* finally started his journey home, exhausted but satisfied. As he walked down a quiet street lined with N flickering streetlights. His programmer brain refused to rest. He stared at the lights and thought, "What if these were elements of an array?"

That stray thought spiraled into a new challenge: What would be the total sum of MEX values for all possible subarrays of this streetlight array?

More formally, given an array of N distinct integers $a_1, a_2, a_3, \dots, a_N$. Your task is to compute:

$$\sum_{1 \leq l \leq r \leq N} \text{MEX}(a_l, a_{l+1}, \dots, a_r)$$

Too tired to solve it himself, *HRK* decided to hand over the puzzle to you.

The *MEX (Minimum Excluded Value)* of an array is the smallest non-negative integer that is not in the array. For example

- The MEX of [2,3,1] is 0 because 0 is not in the array.
- The MEX of [3,1,0,2] is 4 because 0, 1, 2 and 3 are in the array but 4 is not.

Input

The first line contains an integer T ($1 \leq T \leq 10$) — the number of test cases

Each test case consists of two lines:

The first line contains an integer N ($1 \leq N \leq 10^6$) — the total number of streetlights.

The second line contains N integers $a_1, a_2, \dots, a_N$, where ($0 \leq a_i \leq 10^9$), representing the values of the streetlights.

It is guaranteed that all values are distinct, and the sum of N over all test cases does not exceed 10^6 .

Output

For each Test Case output a single integer — total sum of *MEX* of all possible subarrays.

Examples

Input	Output
1 5 0 1 2 3 4	15

F. Rakib's Social Distance Algorithm

Time: 1 s

Memory: 128 MB

Rakib, along with his friends Tajrin, Ahnaf and Nafiz, is traveling by train. As they are best friends, they hope to sit together on this journey.

Each row of the carriage has exactly four seats. Some seats are already taken by members of Rakib's group (we mark those as `#`). Some seats are out of use or reserved and cannot be asked (we mark those as `\*`). The remaining seats are occupied by strangers. Each stranger carries a small integer, a willingness score, which measures how likely they are to agree to swap seats.

Rakib is shy, so he will only ask a stranger to swap if that stranger's willingness score is at least K . He will never approach seats marked `\*`.

Your job is to help Rakib pick a row he and his friends can make entirely theirs by asking strangers with sufficient willingness. For a candidate row to work, every numeric seat must have the *willingness_score* $\geq K$, and the row must contain no `\*`. The number of strangers Rakib must approach equals the number of numeric seats in that row.

We pick the best row by these rules:

- Fewer strangers approaching is better.
- If tied, prefer the row with the larger total willingness sum among those numeric seats.
- If still tied, choose the smallest row index (1-based).

Input

The first line contains two integers N and K .

Each of the next N lines contains 4 tokens separated by spaces. Each token is either #, *, or a non-negative integer.

Constraint

$$1 \leq N \leq 2 \times 10^5$$

$$0 \leq K \leq 2 \times 10^9$$

$$0 \leq \text{willingness_score} \leq 1 \times 10^9$$

Output

If Rakib can claim a row, print two space separated integers:

The chosen row index (1-based) and the number of strangers approached.

Otherwise, print the text "Rag korla?" on a single line without quotes.

Examples

Input	Output
4 10 4 1 # * # # # 10 12 13 14 15 * * * 10	2 1

Input	Output
3 5 # # # # 1 2 3 4 5 6 7 8	1 0
Input	Output
2 10 # # # 9 * * * #	Rag korla?

G. The Corrupted RAID Array

Time: 1 s

Memory: 128 MB

During a server migration, a RAID controller failed, leaving behind a raw data stream of N integers. Technicians suspect the data was stored using Fixed-Block Sectoring. In this architecture, the system chooses a sector size L . It then reads the stream from the very beginning, dividing it into consecutive blocks of exactly L integers. Any integers remaining at the end of the stream that cannot form a full block of length L are discarded as "padding noise".

The Lead Engineer believes the original "Data Packets" are the sums of the integers within these blocks. To reconstruct the file system, you must identify every unique packet sum that could have existed for every possible size L (where $1 \leq L \leq N$).

Once you have identified all possible unique packet sums, you must answer Q queries. Each query provides a capacity X , and you must report how many unique packet sums are less than or equal to X .

Input

- The first line contains N , the number of integers in the stream.
- The second line contains N integers $(a_1, a_2, \dots, a_n)$.
- The third line contains Q , the number of queries.
- The fourth line contains Q integers, each representing a capacity X .

Constraint

$$1 \leq N \leq 10^5$$

$$1 \leq a_i \leq 10^9$$

$$1 \leq Q \leq 10^5$$

$$1 \leq X \leq 10^{18}$$

Output

For each query, output the number of distinct packet sums not exceeding X , each on a new line.

Examples

Input	Output
4 1 2 3 4 3 1 16 4	1 7 4

Notes

The given stream of integers is $A = [1, 2, 3, 4]$ with length $N = 4$. The system evaluates every possible block length $L \in [1, N]$ and extracts full blocks starting from the first element:

- For $L = 1$: The blocks are $[1], [2], [3], [4]$. The sums are $\{1, 2, 3, 4\}$.
- For $L = 2$: The blocks are $[1, 2]$ and $[3, 4]$. The sums are $\{3, 7\}$.

- For $L = 3$: The only full block is $[1, 2, 3]$. The remaining element $[4]$ is ignored. The sum is 6.
- For $L = 4$: The only full block is $[1, 2, 3, 4]$. The sum is 10.

By combining all extracted sums, we get the set of unique packet sums:

$$S = \{1, 2, 3, 4, 6, 7, 10\}.$$

Processing the $Q = 3$ queries:

1. Query $X = 1$: The only sum in $S \leq 1$ is 1. Count = 1.
2. Query $X = 16$: All sums in S are ≤ 16 . Count = 7.
3. Query $X = 4$: The sums $\{1, 2, 3, 4\}$ are ≤ 4 . Count = 4.

H. The Salamanca Sunset

Time: 0.5 s

Memory: 64 MB

Gustavo Fring is a man of order, precision, and efficiency. He is currently monitoring the growth of his regional distribution centers. Gustavo's primary center has an annual revenue of A million dollars, while Hector Salamanca's street-level operation has an annual revenue of B million dollars. Gustavo knows that in this business, you don't just win, you dominate. He considers his operation "perfectly optimized" only when his revenue is exactly double that of Hector's.

Every year, due to market growth, the revenue of both operations increases by exactly 1 million dollars. Gustavo wants to calculate how many years Y must pass until his empire reaches this "perfectly optimized" state. However, Gustavo is also a realist. If it is not possible for this optimization to occur at any point in the future, the window of opportunity has closed and the "Pollos Standard" can never be achieved again. In such cases, output -1.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), the number of test cases. Each test case contains two space-separated integers: A and B ($1 \leq B < A \leq 10^9$).

Output

For each test case, output a single integer Y , representing the number of years after which Gustavo's operation becomes "perfectly optimized". If the optimized state will never occur in the future, output -1.

Examples

Input	Output
3	7
15 4	18
42 12	-1
3 2	

